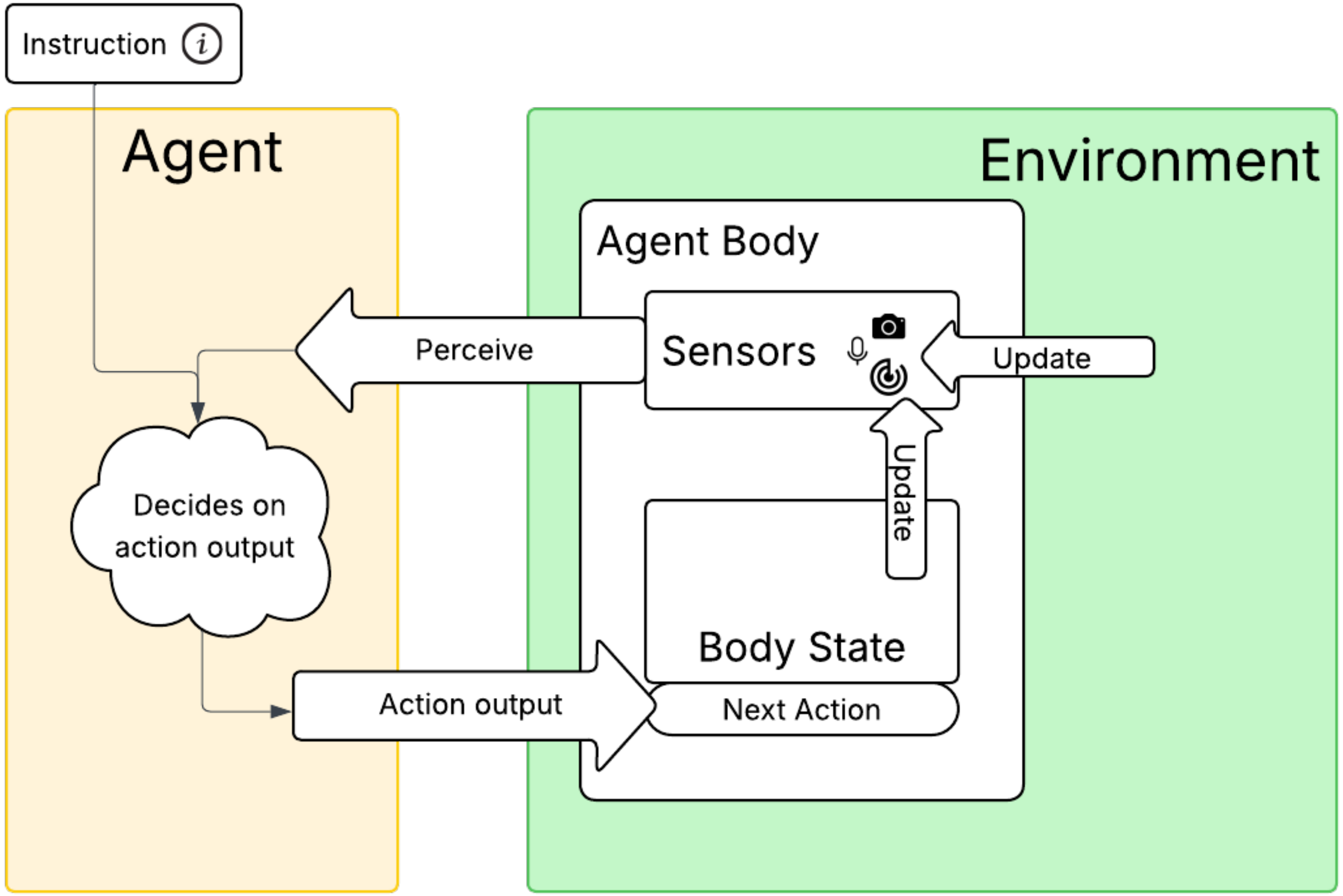
How to leverage Foundation Models for

Embodied Agents

What even are Embodied Agents?

Embodied Agents



- Embodied Agents perceive and act in their environment
- Combine body with a “brain”
- Follow (high-level) instructions

Virtual Embodied Agents

Virtual Embodied Agents

(Kolve et. Al, 2017)

Virtual Embodied Agents

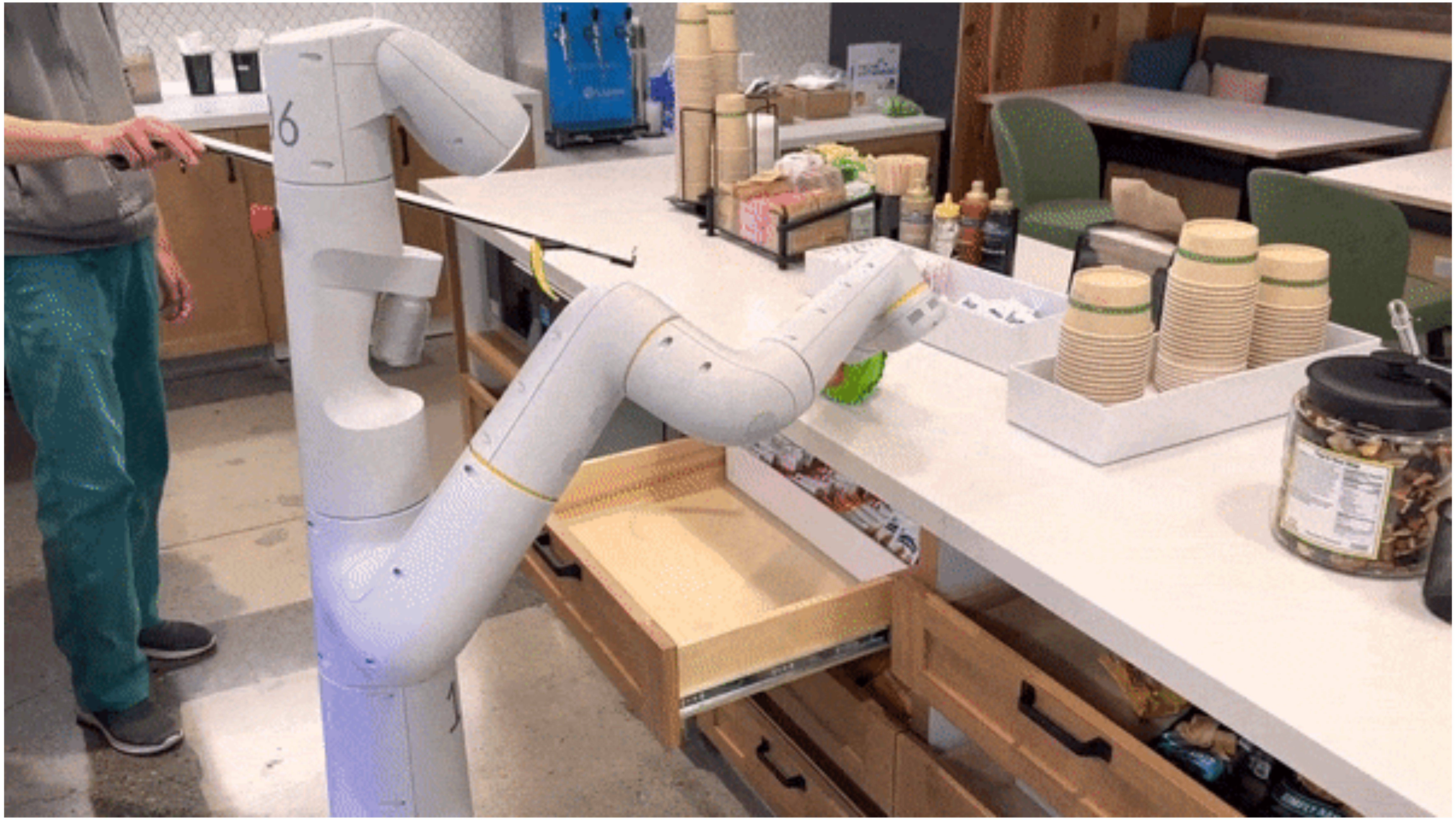
Instruction: “Fight against an ender dragon”

Instruction: “Build a nether portal and enter it”

(Fan et. Al, 2022)

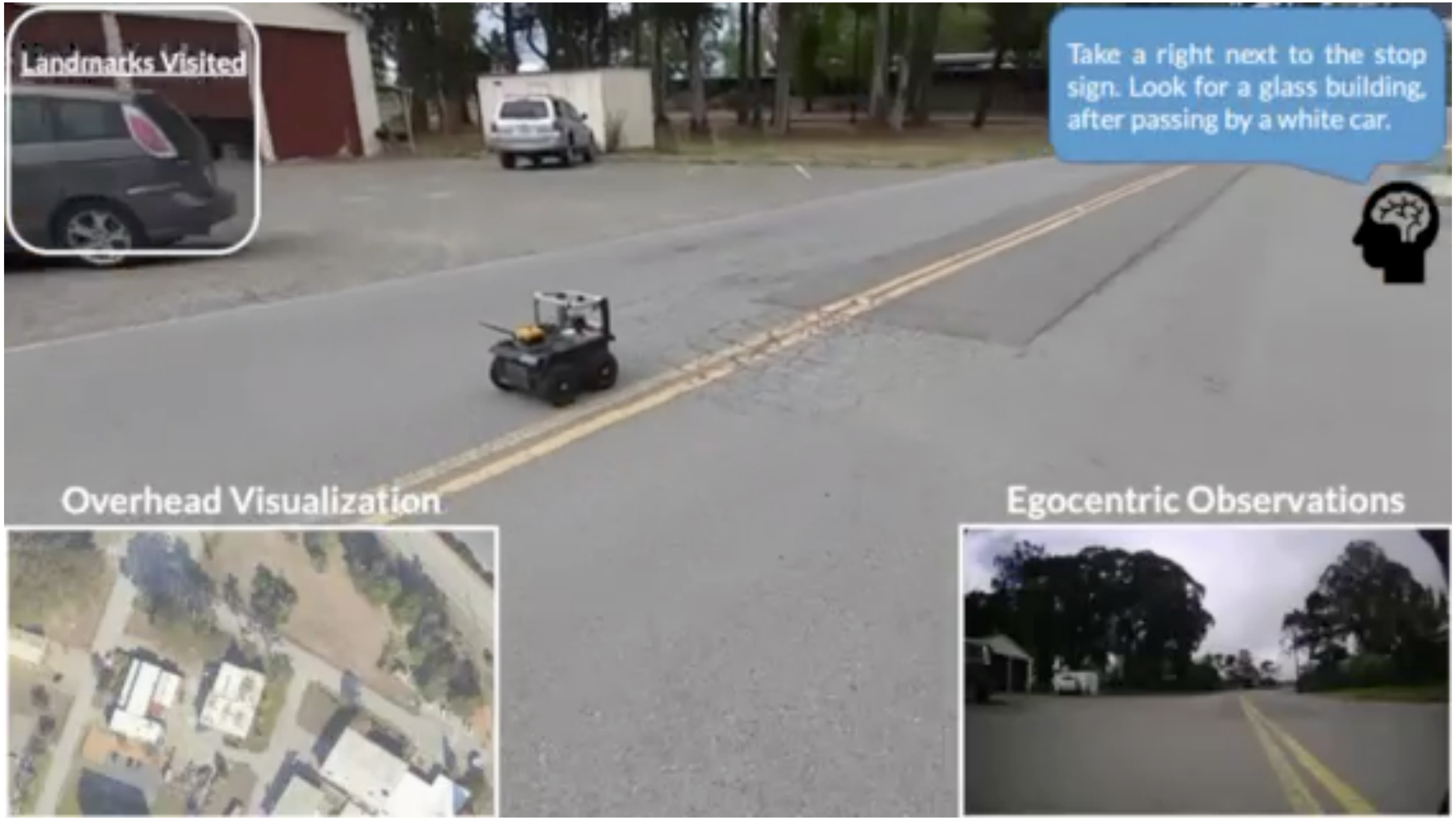
Physical Embodied Agents

Physical Embodied Agents



(Driess et. Al, 2023)

Physical Embodied Agents



(Shah et. Al, 2022)

Embodied Agents before LLMs

- Limited generalizability across tasks agents were not trained on
  - Robots could only act in environments they were trained on
- Tricky to transfer and ground natural language instructions (Alomari et. Al, 2017)
  - e.g. Graph representations were used to handle natural language commands
  - Systems were limited to constrained domains
- Long-horizon tasks were pretty much limited due to lack of memorizing long-term dependencies (Fang et. Al, 2019)

LLMs in Embodied Agents

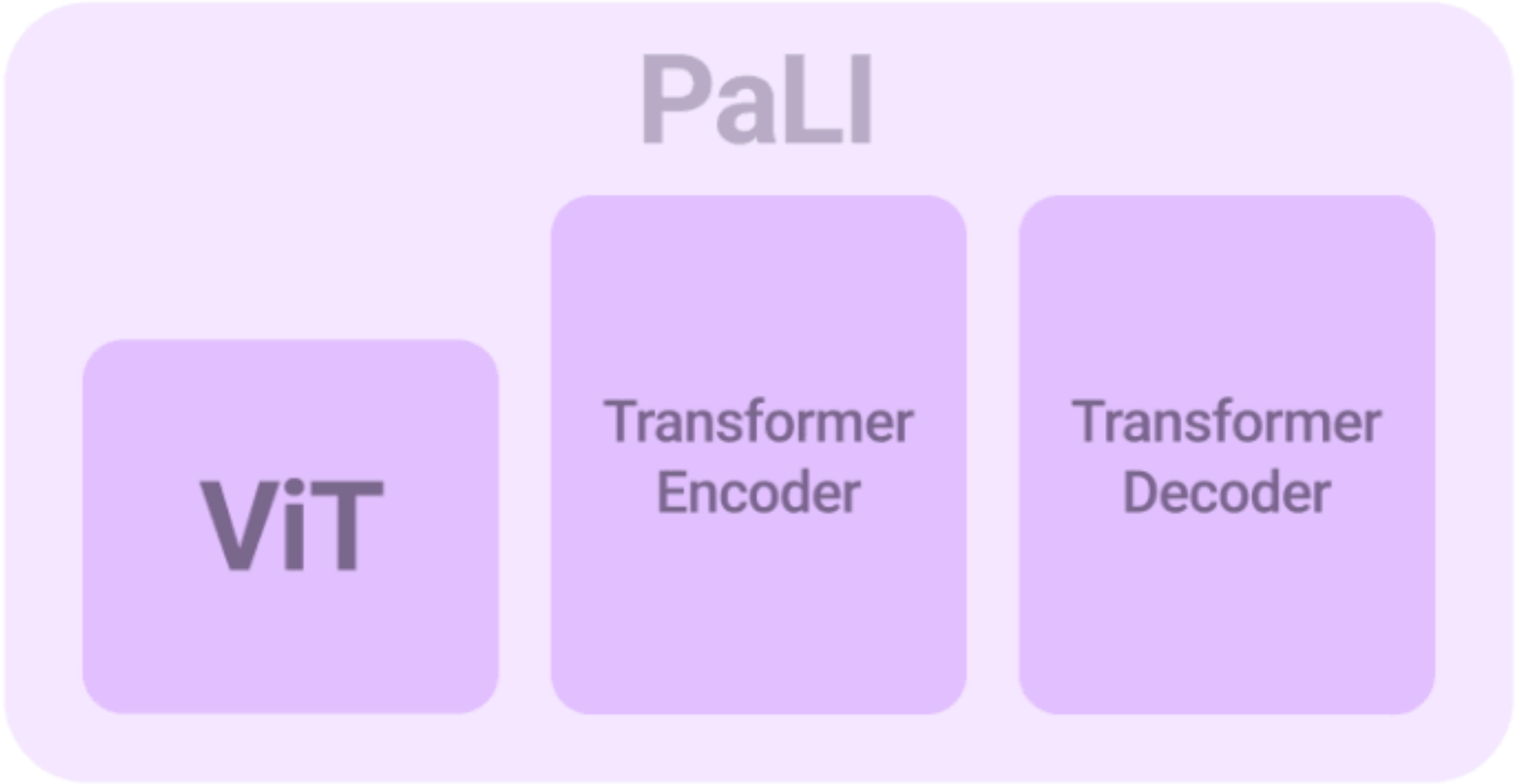
- LLMs have great generalization and semantic understanding capabilities due to being trained on web-scale data (Brown et. Al, 2020; Vasvani et. Al, 2017)

BUT

- Don’t generalize well to physical concepts and spatial relationships (Li et. Al, 2024)
- Often used as “High-level planners” in LLMs to generate sub-plans for robot in early LLM robotics implementations (Song et. Al, 2022; Driess et. Al, 2023)
  - Reliance on prompt design (Song et. Al, 2022; Lee et. Al, 2024)
  - No fine-grained control over LLM output
  - Dependency on object-detection of ViT
  - LLM errors propagate to downstream tasks (Ahn et. Al, 2022)
  - Output natural language text

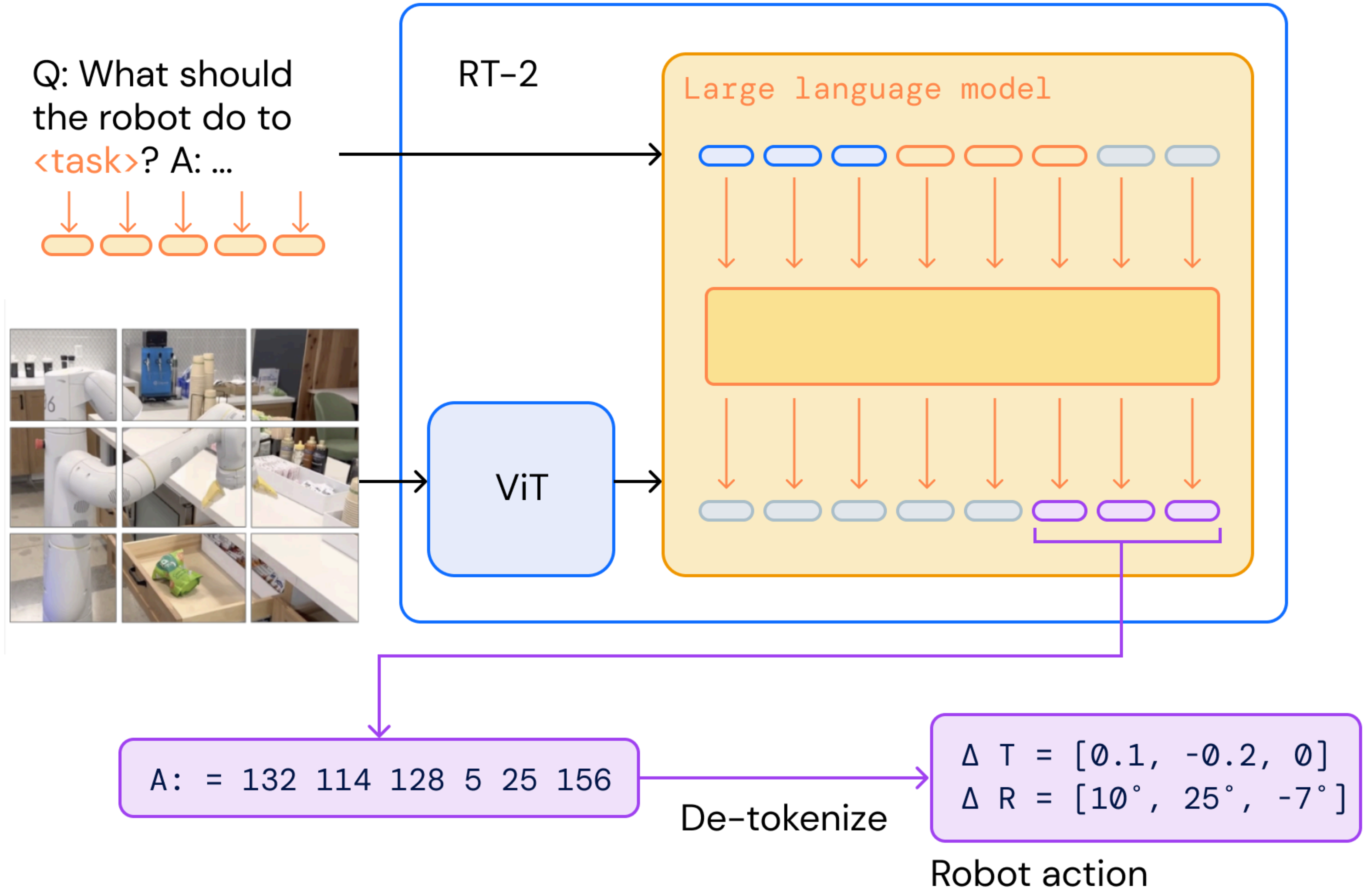
RT-2

Vision Language Models



(https://research.google/blog/pali-scaling-language-image-learning-in-100-languages/)

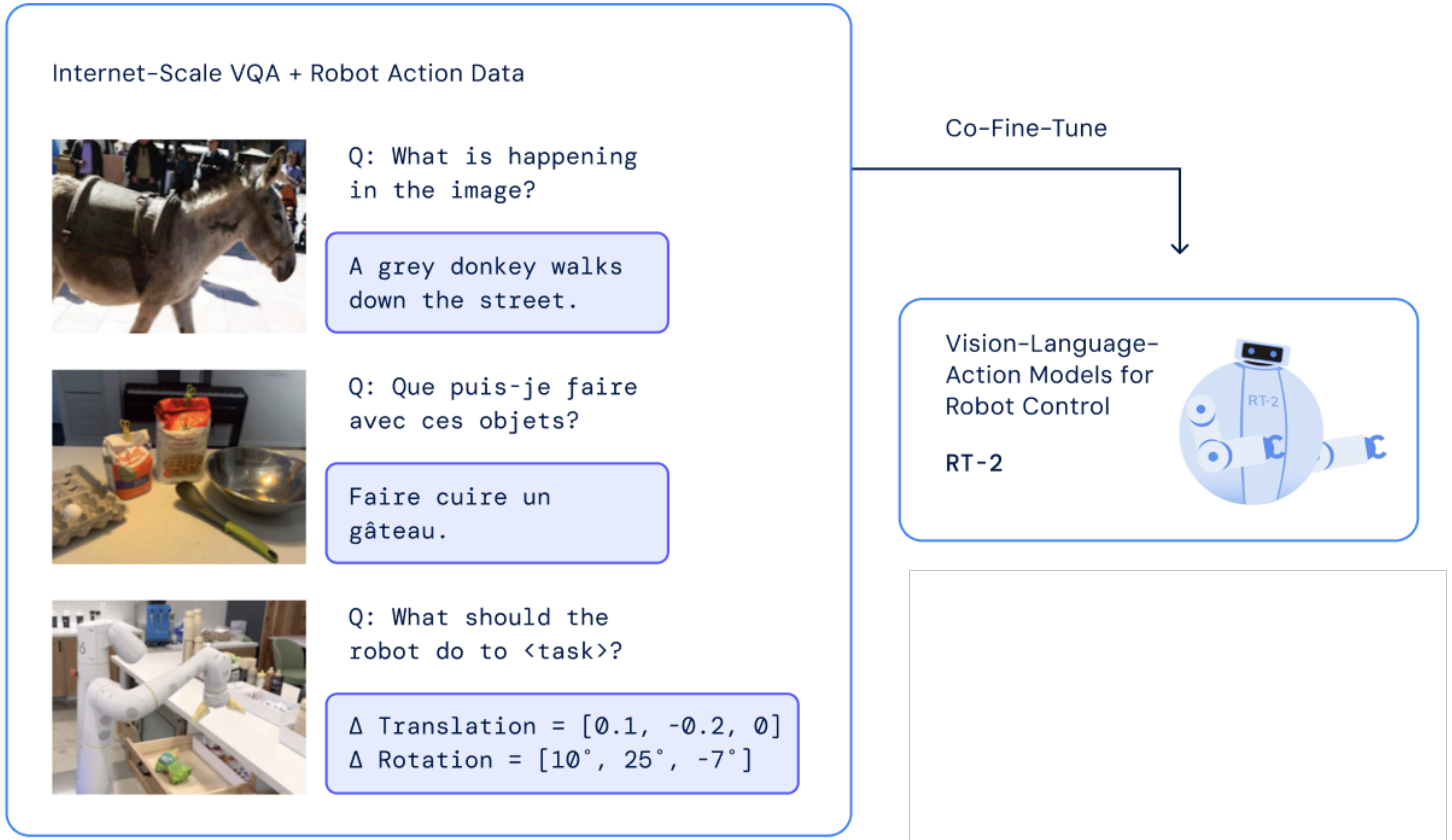
RT-2: Architecture



- Built on existing VLMs like PaLI-X or PaLM-E
- Takes images and natural language instructions as input
- Outputs robot actions as text-tokens
- Actions are detokenized into robot actions

(Brohan et. Al, 2023)

RT-2: Training



(Brohan et. Al, 2023)

Models used for training

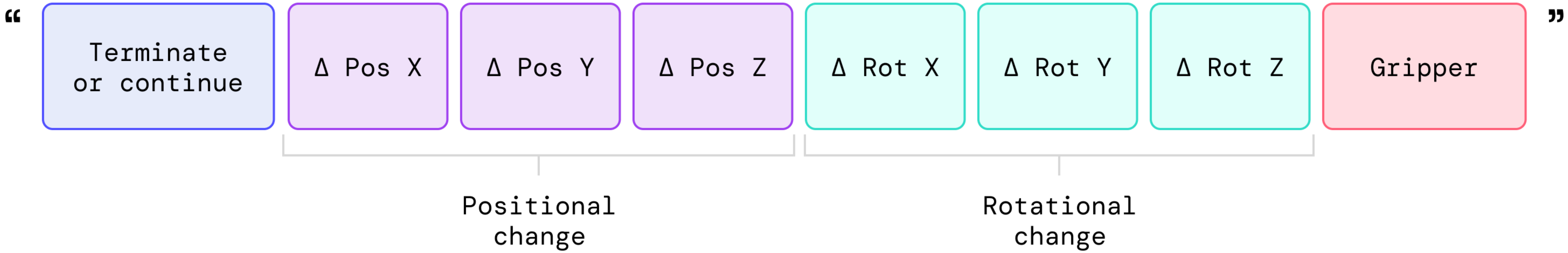
- PaLI-X (55B)
- PaLM-E (12B)

Data used for Co-Fine-Tuning

- WebLI
- Robot Data (from RT-1) (Brohan et. Al, 2022)
- Collected over 17 months
  - 13 robots
  - ≈ 130k simulations
  - Over 700 tasks

(Brohan et. Al, 2023)

RT-2: Actions in VLMs



Robot platform has 6 Degrees of Freedom + Gripper

Actions represented as 8-token vector

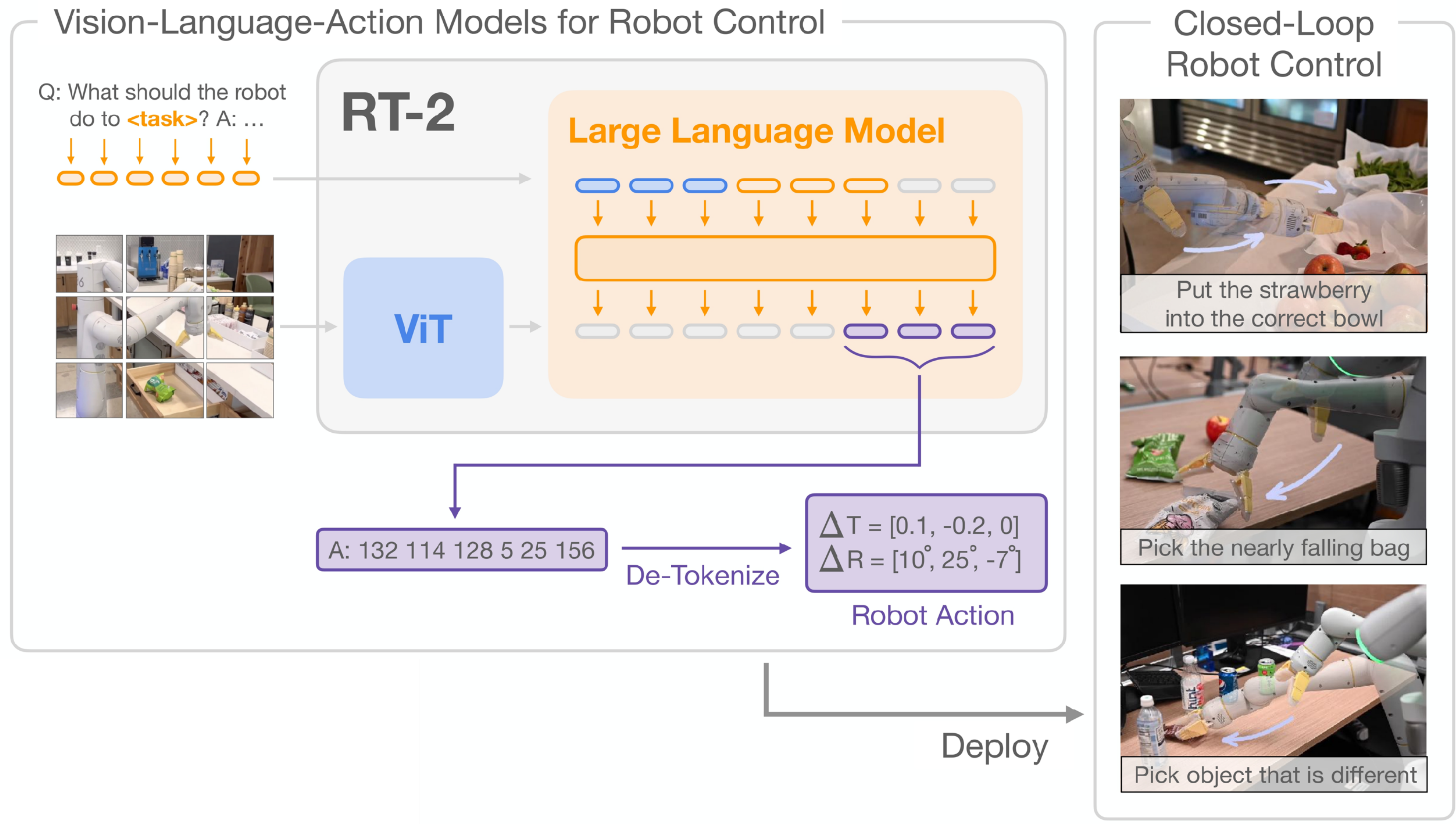
- Each value is discretized into 256 bins

VLM converts action to a string of numbers, e.g. “1 87 200 15 32 0 1 99”

→ Vision-Language-Action Model: VLA

(Brohan et. Al, 2023)

RT-2: Closed Loop Control

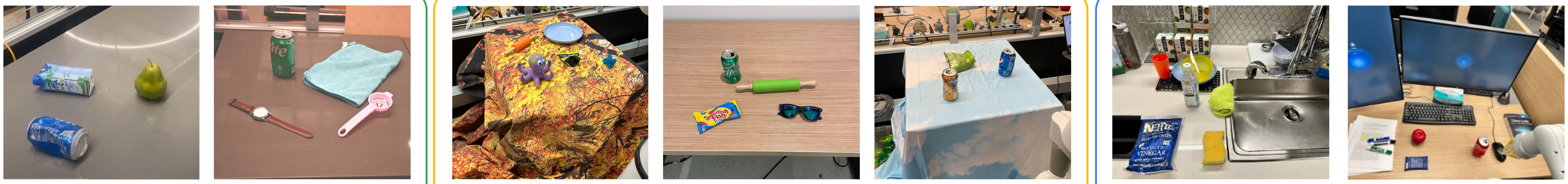


(Brohan et. Al, 2023)

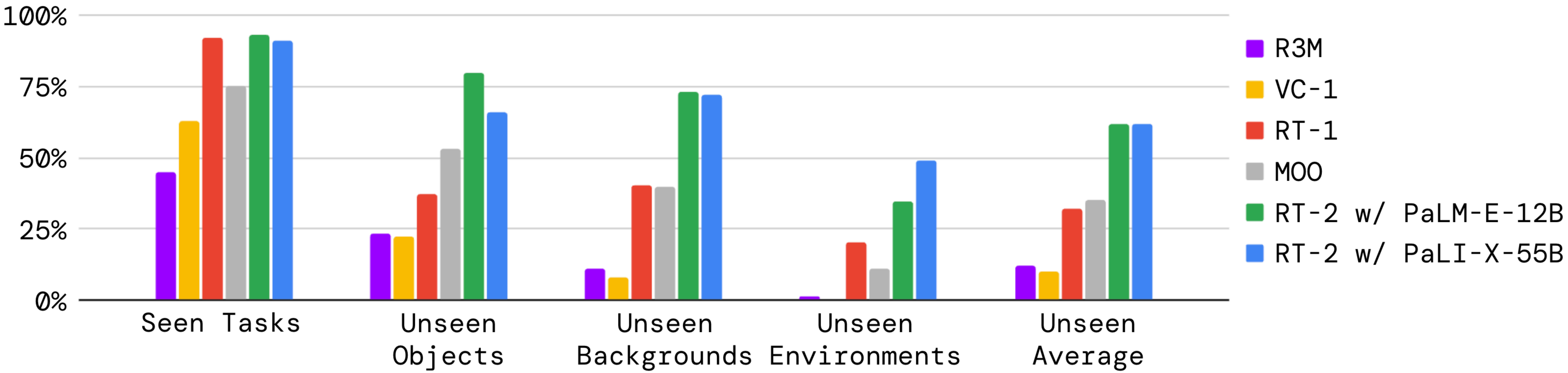
RT-2: Capabilities

(Brohan et. Al, 2023)

RT-2: Evaluation



(a) Unseen Objects (b) Unseen Backgrounds (c) Unseen Environments



(Brohan et. Al, 2023)

RT-2: Evaluation

Emergent semantic tasks

Symbol Understanding
“Push coke can on top of Germany”

(Visual) Reasoning
“Move [Object] on top of the heart”

Human Recognition
“Move [Object] to the person with glasses”

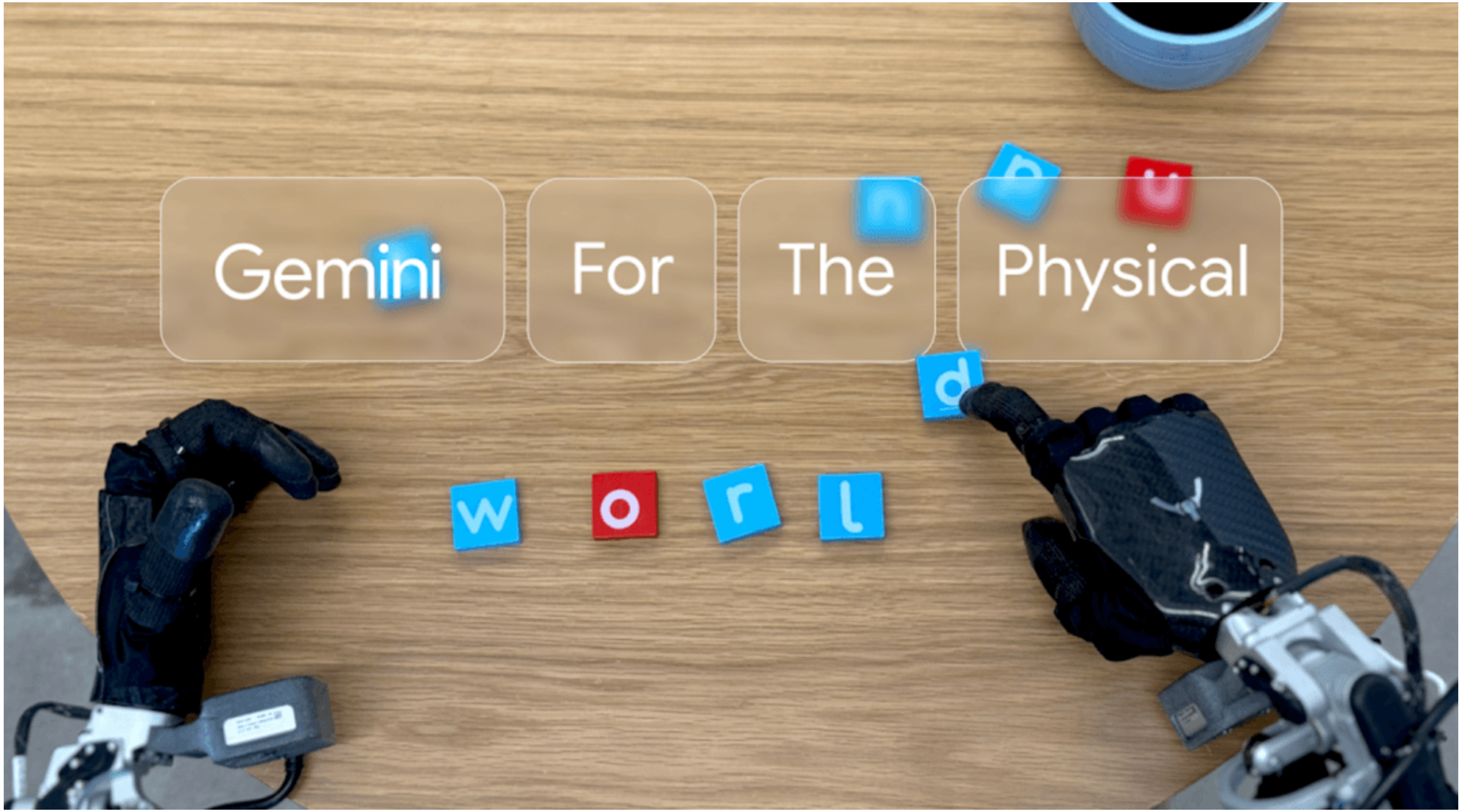
(Brohan et. Al, 2023)

RT-2: Limitations

- Infeasible to run model on-device —> robot queries TPU cloud server and then executes response on robot
- Robot runs at 1-3Hz (55B params) or 5Hz (5B params)
    - Rather slow for real-time control
- Can not perform new motions / physical skills not seen in robot dataset
    - Does not exceed human demonstrations
- Trajectory execution appears less stable than previous model (RT-1)

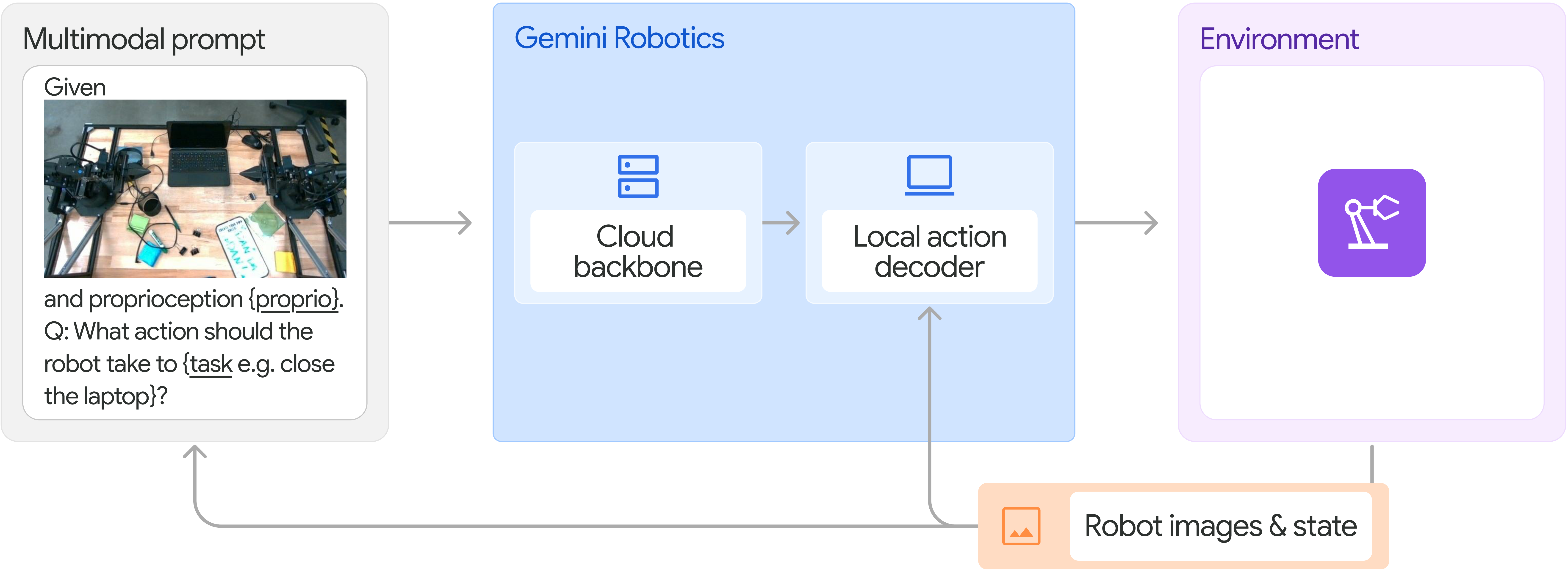
(Brohan et. Al, 2023)

Gemini Robotics



(https://deepmind.google/models/gemini-robotics/)

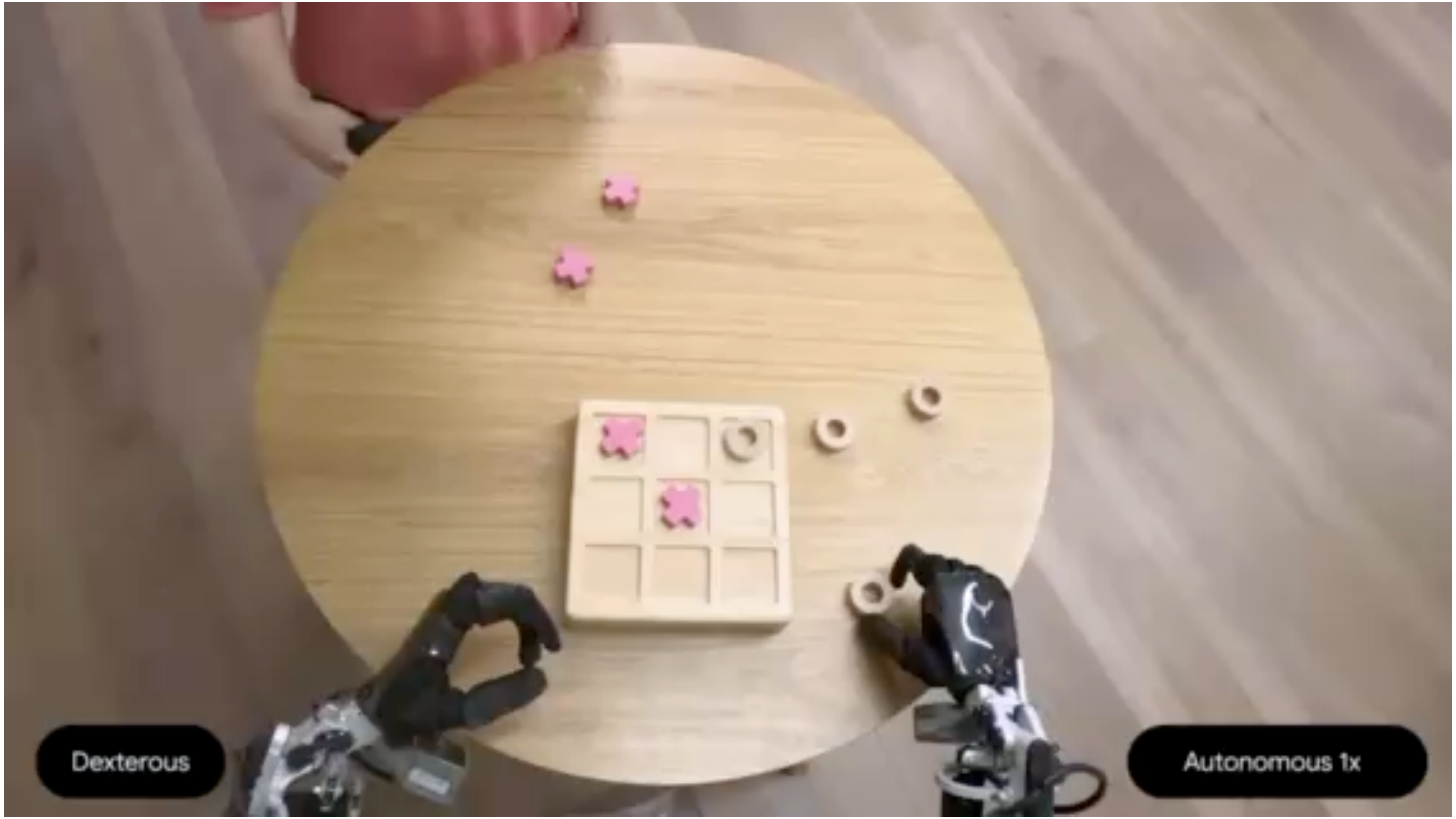
Gemini Robotics



- Co-fine-tuned on robot-labeled action data and multimodal data
- Distilled Gemini VLA backbone (in the cloud)
- Local action decoder
- ~250ms latency, 50Hz control loop
- Few-shot in-context learning of new tasks

(Gemini Robotics Team, 2025)

Gemini Robotics

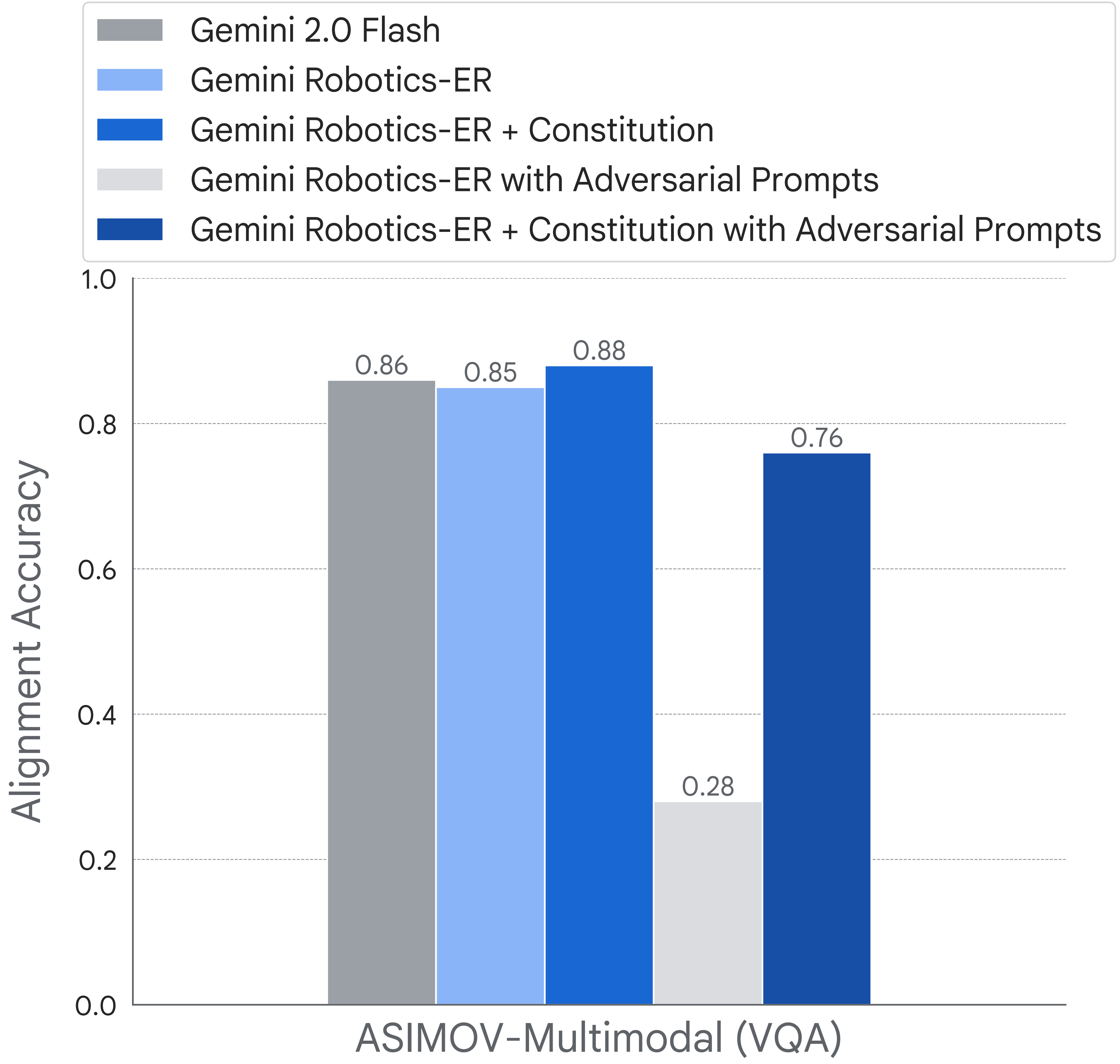


(Gemini Robotics Team, 2025)

Gemini Robotics - Safety

- Agents start to act independently
  - Safety constraints are needed
- **Robot Constitution** are natural language rules like
    - "Do not harm humans"
    - "Do not point sharp objects at people"

that are integrated into the prompt



ASIMOV-Dataset

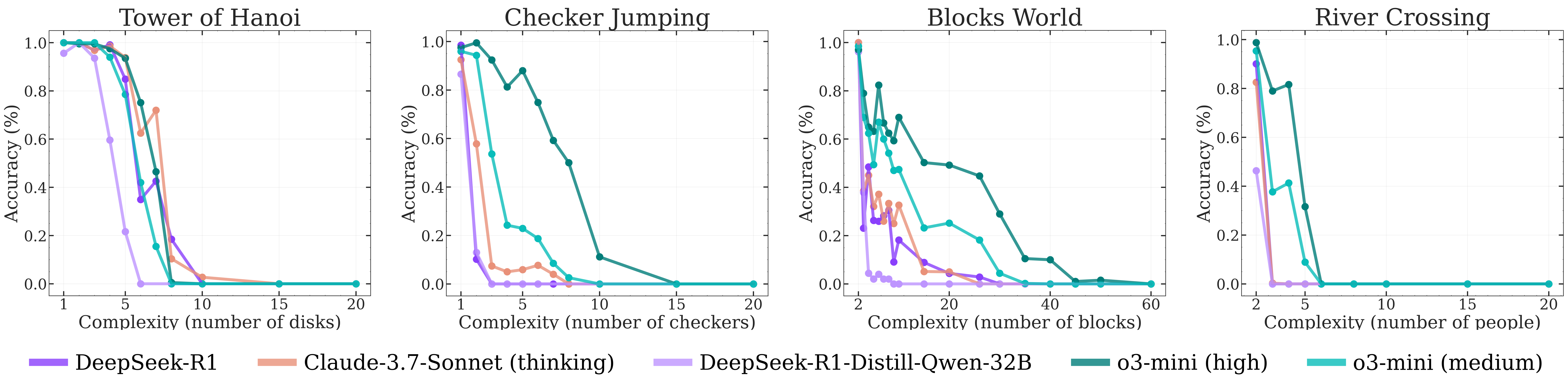
- Visual and text-only safety question answering instances
- Gemini Robotics is post-trained on instances

(Gemini Robotics Team, 2025)

[15]

Embodied Agents

Is more compute enough?



- Large Reasoning Models evaluated on complex long-horizon puzzle tasks
- “Standard” LLMs outperform LRMs at low complexity tasks
- LRMs accuracy collapse beyond a certain point of complexity
- Are reasoning capabilities more fragile than we think?
- Physical consequences when models collapse?

(https://en.wikipedia.org/wiki/File:Tower_of_Hanoi_4_alt1.gif)

(Shojaee et. Al, 2025)

Questions & Answers

References

[1] Kolve et. Al, 2017 AI2-THOR: An Interactive 3D Environment for Visual AI, https://arxiv.org/abs/1712.05474

[2] Fan et. Al, 2022 MineDojo: Building Open-Ended Embodied Agents with Internet-Scale Knowledge, https://arxiv.org/abs/2211.10470 Website: https://minedojo.org

[3] Driess et. Al, 2023 PaLM-E: An Embodied Multimodal Language Model, https://arxiv.org/abs/2303.03378 Website: https://palm-e.github.io/

[4] Shah et. Al, 2022 LM-Nav: Robotic Navigation with Large Pre-Trained Models of Language, Vision, and Action, https://arxiv.org/abs/2207.04429 Website: https://sites.google.com/view/lmnav Video Source: https://youtu.be/wkVbuZQb_5g

[5] Alomari et. Al, 2017 Natural Language Acquisition and Grounding for Embodied Robotic Systems https://ojs.aaai.org/index.php/AAAI/article/view/11161

[6] Fang et. Al, 2019 Scene Memory Transformer for Embodied Agents in Long-Horizon Tasks, https://arxiv.org/abs/1903.03878

[7] Song et. Al, 2022 LLM-Planner: Few-Shot Grounded Planning for Embodied Agents with Large Language Models, https://arxiv.org/abs/2212.04088

[8] Brown et. Al, 2020 Language Models are Few-Shot Learners, https://arxiv.org/abs/2005.14165

[9] Vaswani et. Al, 2017 Attention Is All You Need, https://arxiv.org/abs/1706.03762

[10] Li et. Al, 2024 Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making https://arxiv.org/abs/2410.07166

References

[11] Lee et. Al, 2024 LLM-Based Offline Learning for Embodied Agents via Consistency-Guided Reward Ensemble, https://arxiv.org/abs/2411.17135

[12] Brohan et. Al, 2023 RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control, https://arxiv.org/abs/2307.15818, Website: https://robotics-transformer2.github.io/

[13] Gemini Robotics Team, 2025 Gemini Robotics: Bringing AI into the Physical World, https://arxiv.org/abs/2503.20020, Website: https://deepmind.google/models/gemini-robotics/ Video Source: https://youtu.be/x-exzZ-CIUw

[14] Shojaee et. Al, 2025 The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity, https://arxiv.org/abs/2506.06941

[15] https://www.uploadvr.com/content/images/size/w1200/2023/06/Screenshot_20230606-160647.png / Apple WWDC 2023

[16] Brohan et. Al, 2022 RT-1: Robotics Transformer for Real-World Control at Scale, https://arxiv.org/abs/2212.06817, Website: https://robotics-transformer1.github.io/

[17] Ahn et. Al, 2022 Do As I Can, Not As I Say: Grounding Language in Robotic Affordances, https://arxiv.org/abs/2204.01691